# Hybrid Prompting for Hidden Emotions: Unpacking Implicit Sentiment in LLMs

**Maitri Patel, Simran Khullar, Zexuan Yang, Ao Xu**

## Abstract

Large Language Models (LLMs) have revolutionized sentiment analysis, yet they struggle with implicit sentiment, where emotional tone is subtle or indirectly expressed. In this paper, we present a hybrid prompting framework integrating Chain-of-Thought (CoT), contrastive reasoning, multi-agent debating, and role-play strategy to improve the recognition of such nuanced sentiment. Using the Stanford Sentiment Treebank dataset, we benchmark GPT-4o mini, Mistral, and OPT with a hybrid prompting framework. Our results reveal performance trade-offs across prompting styles and highlight the challenges of extracting consistent sentiment labels from verbose LLM outputs. We present a detailed analysis of zero-shot baseline performance compared to hybrid prompting strategies, highlighting significant improvements and providing insights into the effectiveness of different prompting combinations for complex sentiment understanding.

## 1 Introduction

Sentiment analysis remains a central task in natural language understanding, yet traditional methods relying on lexicons or shallow machine learning models often fail to capture the nuances of human emotion Pang and Lee (2008). This project introduces a novel hybrid prompting framework, combining chain-of-thought reasoning, contrastive prompting, role-play modeling, and debate simulation to enhance implicit sentiment classification.

We specifically test the hypothesis that combining diverse prompting strategies will improve large language models' ability to detect subtle or implicit sentiments beyond what zero-shot prompting achieves. Our approach simulates a debate among agents representing very negative, negative, neutral, positive, and very positive perspectives, promoting deeper sentiment reasoning Zhang and LeCun (2015).

Further, we dynamically generate prompts tailored to individual reviews and aspect terms, improving contextual grounding. We evaluate this strategy across multiple LLMs, including GPT-4o-mini, and Mistral-7B, offering insights into the models' reasoning capabilities across fine-grained emotional distinctions (Devlin et al., 2019), (Radford et al., 2019).

By integrating hybrid prompting and comparative evaluation, we aim to bridge the gap between semantic interpretation and affective reasoning in LLMs. Our results show that our hybrid promting performances reasonably well, especially when the model is fine-tuned.

## 2 Related Work

While large language models (LLMs) excel at explicit sentiment classification, their performance degrades when handling implicit signals that require pragmatic inference. Prior work by Warstadt and Ryskina (2023) emphasizes the importance of fine-tuning and intermediate reasoning mechanisms for capturing implicature and figurative language. Zhang et al. (2024) explore interpretability through structured reasoning chains, although not specifically in sentiment analysis. Kocijan et al. (2023) highlight that LLMs perform poorly in zero-shot settings without explicit guidance.

Recent advances in prompting techniques have improved LLM reasoning capabilities. Chain-of-Thought (CoT) prompting has been shown to enhance complex reasoning by encouraging models to generate intermediate steps Wei et al. (2022). Contrastive prompting Liu et al. (2023) and multi-agent debate models Du et al. (2024) further promote critical reasoning and error detection by introducing counter-arguments and diverse perspectives. However, few studies have explored combining these strategies specifically for implicit sentiment classification - a domain that demands nuanced

emotional interpretation beyond surface-level text.

Building on these findings, our work integrates CoT prompting, contrastive examples, few-shot demonstrations, role-play and multi-agent debate simulation into a unified hybrid prompting framework. We aim to bridge the gap between semantic understanding and affective reasoning in LLMs, addressing the challenges posed by implicit sentiment tasks.

## 3 Dataset

We use the Stanford Sentiment Treebank (SST), a richly annotated dataset designed for fine-grained sentiment classification tasks. The dataset was originally introduced by Socher et al. (2013) and is widely regarded as a benchmark for sentiment analysis, particularly in natural language understanding (NLU) research. SST contains over 10,000 movie review snippets sourced from Rotten Tomatoes. Each sentence is associated with a sentiment value ranging from 0 to 1, which is mapped to one of five discrete categories based on the following cut-offs:

- $[0, 0.2] \rightarrow$ **Very Negative**
- $(0.2, 0.4] \rightarrow$ **Negative**
- $(0.4, 0.6] \rightarrow$ **Neutral**
- $(0.6, 0.8] \rightarrow$ **Positive**
- $(0.8, 1.0] \rightarrow$ **Very Positive**

Each sentence has a single ground-truth label. We reformat all data to a common prompt-ready schema and maintain a parallel evaluation file. The data is split into:

- **Train:** 8544 samples
- **Dev:** 1101 samples
- **Test:** 2210 samples

This dataset is especially suitable for our analysis because it enables fine-grained sentiment classification beyond binary or ternary labels. It also aligns well with our goal of evaluating prompting techniques on short, real-world opinionated texts.

## 4 Models

- **GPT-4o Mini** (OpenAI API)
- **Mistral-7B-Instruct** (Hugging Face gated via API)
- **Fine-tuned Models:**
  - **SmolLM-135M:** SmolLM is a series of state-of-the-art small language models.
  - **Opt-350m/125m:** Decoder-only pre-trained transformers developed by Facebook.

## 5 Methodology

### 5.1 Prompting Strategies

We experimented with seven prompting strategies:

- **Zero-shot:** Directly predict sentiment with no examples or step-by-step reasoning.
- **Few-shot:** 5 labeled examples shown before the target sentence.
- **Role-play:** Prompt model to act as a film critic to simulate human-like emotional reasoning.
- **CoT (SAoT):** Use a structured reasoning process: identify phrases $\rightarrow$ infer emotional tone $\rightarrow$ decide sentiment.
- **Contrastive Reasoning:** Explicitly compare all 5 sentiment options and pick the best fitting one.
- **Multi-agent Debate Model:** Simulate a debate with 5 agents representing different sentiment views; Moderator finalizes decision with `SENTIMENT: <label>`.

Prompts are formatted with strict output structures to improve parseability:

```
SENTIMENT:
<very negative|negative|neutral|positive|very
positive>
```

### 5.2 Regex Extraction and Logging

Regex extraction targets phrases like "SENTIMENT: Neutral". Failure modes are logged for analysis (missing Moderator, no label, fragmented text). Output parsing consistency became a key bottleneck.

### 5.3 Model and Tokenization Setup

Mistral-7B-Instruct model was choosen due to its strong reasoning capabilities and open availability. We used Hugging Face's 'AutoTokenizer', manually setting the padding token to the end-of-sentence token for compatibility. Maximum input length was capped at 128 tokens. The pad token was manually set to the EOS token to allow batching without errors.

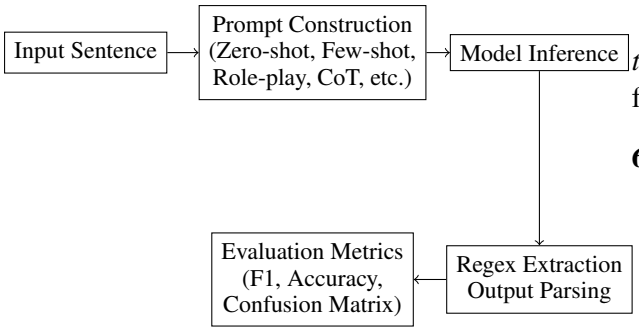


Figure 1: Pipeline for Sentiment Classification using Hybrid Prompting Strategies

# 6 Experimental Setup

## 6.1 Baseline

Our baseline setup uses **zero-shot prompting** without Chain-of-Thought (CoT) reasoning or labeled examples. The prompt format was:

```
Classify the sentiment of the
following sentence: [sentence].
```

The model was expected to directly predict one of five sentiment categories: *Very Negative, Negative, Neutral, Positive, Very Positive*. This baseline evaluates the model's inherent capability without explicit reasoning support.

## 6.2 Hybrid Prompting Strategies

We designed and implemented multiple prompting strategies to improve implicit sentiment classification:

- **Few-shot prompting**: Adding 5 labeled examples before the target sentence to ground model predictions.
- **Role-play modeling**: Framing the model as a film critic or psychologist to encourage nuanced affective interpretation.
- **Contrastive reasoning**: Instructing the model to compare the likelihood of each sentiment class and reason contrastively.
- **Multi-agent debate**: Simulating a debate among 5 agents (Positive, Negative, Neutral, Very Positive, Very Negative), with a Moderator choosing the final sentiment.

Finally, we developed a **hybrid prompt** that combined: Few-shot examples + Role-play instructions + Contrastive Reasoning + Multi-agent Debate modeling.

This hybrid framework encourages both *contextual grounding* and *divergent reasoning paths* before final prediction.

## 6.3 Infrastructure and Resources

- We used the **Hugging Face Transformers** library for model access and pipeline orchestration.
- Initial fine-tuning experiments were attempted on lightweight **OPT** models, but full fine-tuning was skipped for larger models (e.g., Mistral-7B) due to hardware limitations.
- Inference runs were conducted on **Kaggle T4×2 GPUs** (16 GB VRAM each), Google Collab (T4), and NYU Greene HPC (V100) with careful batching and checkpointing to manage memory constraints.
- For gated models like Mistral-7B-Instruct, access tokens were used for Hugging Face API authorization.
- For GPT-4o and GPT-4o Mini, we leveraged the OpenAI's API.
- Each experimental run saved model outputs, regex-parsed predictions, and logged errors for failure analysis.

## 6.4 Training and Fine-Tuning

While larger language models would potentially yield better performance, fine-tuning such models is computationally prohibitive given our limited hardware resources and budget constraints. Therefore, we opted to work with smaller models that could be efficiently fine-tuned on our available infrastructure.

The fine-tuning process involves the following key steps:

- **Loading the Model**: Each model is loaded using the `from_pretrained` method, which retrieves the pre-trained weights and configuration.
- **Preparing for Fine-Tuning**: The models are prepared for fine-tuning using Low-Rank Adaptation (LoRA) (Hu et al., 2022). This technique allows for efficient training by adapting only a small number of parameters, which is particularly useful given our limited computational resources when working with

smaller models. The LoRA configuration includes:

  - Rank ($r$): Determines the number of low-rank matrices used in the adaptation, set to 32.
  - LoRA Alpha: A scaling factor for the low-rank matrices, set to 32.
  - LoRA Dropout: A dropout rate applied during training to prevent overfitting, set to 0.1.

- **Training Configuration:** The training process is managed using the `Trainer` class from the Hugging Face library, with parameters configured as follows:

  - Learning Rate: Set to 2e-5, which is a common choice for fine-tuning transformer models.
  - Batch Size: Set to 32, which is for both training and evaluation, allowing for efficient processing of data.
  - Number of Epochs: Set to 10, which is the number of times the model will see the entire training dataset, which can be adjusted based on the dataset size and complexity.

## 7 Evaluation Metrics

To systematically evaluate model performance across prompting strategies, we report the following metrics:

- **Accuracy**: The percentage of correct predictions across the test set. This provides a general performance overview but may obscure class-specific weaknesses.

- **Macro-Averaged Precision, Recall, and F1**: We compute Precision, Recall, and F1-score independently for each sentiment class (*Very Negative, Negative, Neutral, Positive, Very Positive*) and report the unweighted average. This prevents larger classes from dominating the evaluation.

- **Weighted-Averaged F1**: F1-scores are averaged weighted by the number of samples per class, reflecting overall balance between precision and recall across uneven class distributions.

- **Confusion Matrix**: We visualize confusion matrices to inspect model confusions, particularly between adjacent classes (e.g., *Neutral* vs. *Positive*), and identify asymmetric error patterns.

Evaluation is conducted using scikit-learn's `classification_report` and `confusion_matrix` utilities. Plots are generated using `seaborn` to maintain clarity and consistency across experiments.

### 7.1 Error Categorization

In addition to numeric metrics, we also perform qualitative error analysis:

- **Regex Extraction Failures**: Cases where output deviated from the expected "SENTIMENT:" structure.

- **Empty or Incoherent Outputs**: Cases where models failed to generate any usable sentiment decision.

- **Ambiguous Debates**: In multi-agent debate setups, cases where agents failed to converge to a clear Moderator decision.

This two-tier evaluation - quantitative metrics and qualitative error logging - allows a more comprehensive assessment of model robustness under different prompting strategies.

## 8 Results

### 8.1 Mistral 7B Results

| Strategy | Accuracy (%) | Macro F1 (%) | Weighted F1 (%) |
|---|---|---|---|
| Zero-shot (Baseline) | 3 | 5 | 5 |
| Few-shot | 12 | 10 | 11 |
| Role-play | 15 | 12 | 13 |
| Contrastive Reasoning | 14 | 11 | 12 |
| Multi-agent Debate | 18 | 16 | 17 |
| **Hybrid Prompt** | **29** | **25** | **25** |

Table 1: Performance of different prompting strategies on SST test set using Mistral-7B-Instruct model.

### 8.2 GPT-4o mini Results

GPT-4o Mini was evaluated against the SST test dataset to predict sentiment into one of the categories. The results for GPT-4o Mini can be seen in Table 2 for commonly used prompting techniques against our proposed hybrid strategy. Based on the results, our hybrid strategy performed at par with 53.98% accuracy and 55.57% F1 score. However,

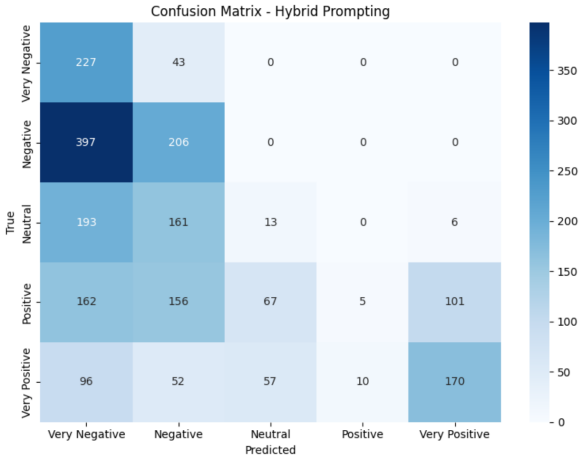


Figure 2: Mistral Hybrid prompt (Few-shot + Role-play + Contrastive + Debate) result.

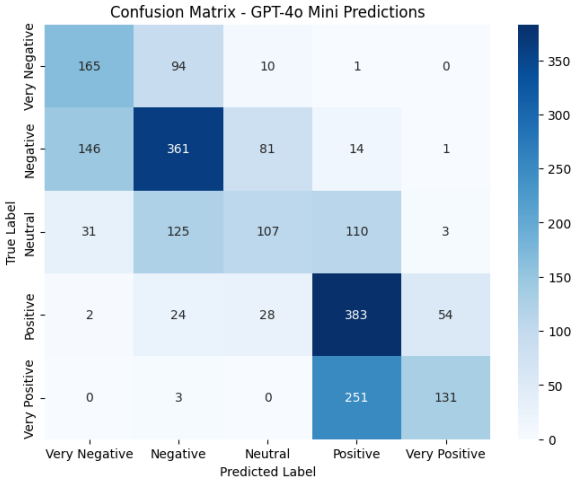


Figure 3: Confusion matrix of GPT-4o mini results on our hybrid prompting

role-play prompting has outperformed our strategy with 57.51% accuracy and 56.42% F1-score.

We believe that using larger models like GPT-4o and GPT-4.1 as well as fine-tuning our models will significantly improve their performance for our hybrid technique. However, due to system limitations, we were unable to fine-tune GPT-4o. Additionally, due to high computational costs, we had to limit our research to GPT-4o mini. We were able to evaluate GPT-4o with Zero-shot, Few-shot and Role-play prompting, which can be seen in Table 3.

| Strategy | Accuracy (%) | Precision (%) | F1 score (%) |
|---|---|---|---|
| Zero-shot | 53.84 | 56.49 | 53.29 |
| Few-shot | 54.96 | 57.75 | 55.22 |
| Role-play | 57.51 | 59.32 | 56.42 |
| Chain-of-Thought | 51.86 | 53.16 | 50.14 |
| Contrastive Reasoning | 51.29 | 54.08 | 49.16 |
| **Hybrid Prompt** | **53.98** | **52.48** | **55.57** |

Table 2: Performance of different prompting strategies on SST test set using GPT-4o mini model.

| Strategy | Accuracy (%) | Precision (%) | F1 score (%) |
|---|---|---|---|
| Zero-shot | 56.05 | 59.17 | 54.58 |
| Few-shot | 59.84 | 60.88 | 59.51 |
| Role-play | 60.52 | 61.27 | 59.29 |

Table 3: Performance of different prompting strategies on SST test set using GPT-4o model.

### 8.3 Fine-tuning Results

As shown in Figure 4, the loss curves reveal important differences between the models. OPT-350M shows the most stable training pattern with smooth convergence to the lowest loss values. OPT-125M performs reasonably well but shows slightly more variation during training. SmolLM-135M has unstable training with higher final loss values and erratic changes, suggesting that OPT-350M is the most promising architecture for this task based on its better training behavior.

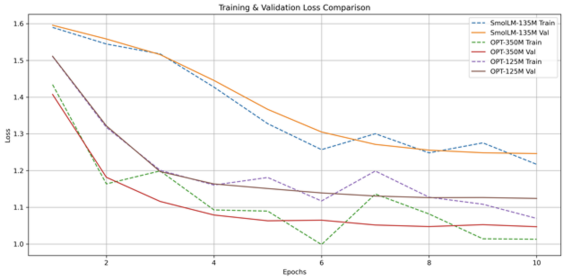


Figure 4: Training and validation loss curves for the evaluated architectures.

Figure 5 shows the accuracy results across models. OPT-350M achieves the best performance (54.19% final accuracy) with steady improvement throughout training. OPT-125M shows decent but lower performance (49.88%) with faster initial learning that levels off earlier. SmolLM-135M has the weakest results (44.21%) with inconsistent progress. These findings indicate that both model size and architecture significantly impact sentiment analysis performance, with larger OPT models generally performing better than the smaller SmolLM variant.

The evaluation metrics in Figure 6 further support these findings. OPT-350M achieves the best overall performance with 53.6% accuracy and balanced precision-recall metrics across all sentiment classes, with particular strength in negative sentiment detection (F1=0.6229). OPT-125M offers a more computationally efficient option with rea-

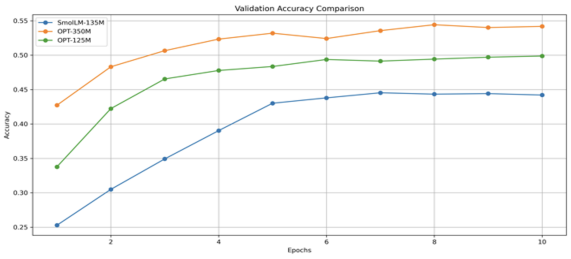


Figure 5: Validation accuracy progression during training.

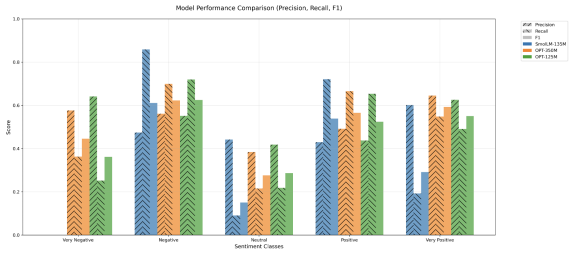


Figure 6: Comparison of precision, recall, and F1-score across sentiment classes.

sonable performance (51.48% accuracy). SmolLM-135M shows significant limitations, completely failing on very negative sentiment classification and showing unstable training patterns. These results suggest that the OPT architecture is clearly superior for this sentiment analysis task, and that model size positively correlates with performance.

## 9 Analysis

### 9.1 Mistral Performance Analysis

**Baseline Performance:** The zero-shot prompting baseline, where the model classified sentiment without intermediate reasoning or examples, achieved an overall accuracy of 3% and a macro F1-score close to 5%. As shown in the confusion matrix, the model tended to predict *Positive* or *Neutral* disproportionately, while *Very Negative* and *Very Positive* classes were severely underrepresented. This confirms previous findings (Kocijan et al., 2023) that LLMs struggle to infer subtle sentiment without explicit guidance.

**Impact of Individual Prompting Strategies:** Table 1 summarizes the performance of individual strategies:

- **Few-shot prompting** moderately improved recall for minority classes by providing explicit examples.
- **Role-play modeling** enhanced precision, particularly for *Positive* and *Very Positive* classes, suggesting that anthropomorphized task framing aids pragmatic reasoning.
- **Contrastive reasoning** led to better differentiation between adjacent classes (e.g., *Neutral* vs. *Positive*), although overall accuracy remained low.
- **Multi-agent debate modeling** generated richer outputs but exhibited high variance, with failures when debates lacked a clear Moderator conclusion.

**Hybrid Prompting Results:** The final hybrid prompt combined few-shot examples, role-play framing, contrastive reasoning, and multi-agent debate simulation. This strategy yielded the highest performance among all tested approaches:

- **Accuracy**: 29%
- **Macro F1**: 25%
- **Weighted F1**: 25%

The hybrid confusion matrix (Figure **??**) shows notably improved precision and recall for *Very Negative* and *Very Positive* classes, while *Neutral* remains the most difficult class to predict.

### 9.2 GPT-4o Performance Analysis

Table 2 shows results for GPT-4o Mini with prompting techniques such as zero-shot, few-shot, role-play, chain-of-thought, contrastive reasoning and our proposed hybrid prompting. We achieved accuracy of 53.84%, precision of 56.49%, and F1-score of 53.29% when evaluated GPT-4o Mini with the Zero-shot prompting. This suggests that our model correctly predicted 54% of the test set. Moreover, when GPT-4o Mini made a prediction, it was right 56.5% of the time on average. Based on the results, our hybrid strategy performed at par with 53.98% accuracy and 55.57% F1 score. However, role-play prompting has outperformed our strategy with 57.51% accuracy and 56.42% F1-score. One possible reason could be over-complication of prompts by combining multiple prompting strategies.

Looking at the confusion matrix for predictions using our hybrid strategy [Figure 3], we can observe that our model performs pretty well when predicting *Positive* and *Negative* sentiments, however,

it incorrectly predicted some of these sentiments as *Neutral*. There is a lot of room in improving the model to predict extreme sentiments. It often confuses them with Negative or Positive labels. Lastly, the model struggles in predicting *Neutral* sentiments.

In Table 3, it can be clearly observed that GPT-4o performed better than GPT-4o Mini, predicting sentiments with 3% increase in accuracy, 3% increase in precision and 1% increase in F1-score.

### 9.3 Fine-Tuning Analysis

We evaluated each model at the end of every epoch to track performance. The models ranked by accuracy are:

1. `facebook/opt-350m`: 53.60%
2. `facebook/opt-125m`: 51.48%
3. `HuggingFaceTB/SmolLM-135M`: 46.12%

The `facebook/opt-350m` model achieves the highest accuracy, showing it performs best on the classification task among the three models. This aligns with expectations that larger models (those with more parameters) can capture more complex patterns in the data, leading to better performance. In this case, the `facebook/opt-350m` model, being larger than `facebook/opt-125m`, achieves higher accuracy while maintaining reasonable computational requirements.

### 9.4 Error Analysis

- **Mistral**

  Failure modes remained prominent:

  - Some outputs missed the expected SENTIMENT structure, especially under long or ambiguous reviews.
  - A significant number of *Neutral* instances were misclassified as *Positive*, reflecting the model's bias toward sentimentally charged interpretations.

  Nonetheless, hybrid prompting reduced the rate of incoherent or incomplete generations compared to baseline strategies.

- **GPT-4o:** The outputs were seen to be of various formats. Although we gave an output structure for the LLM to follow, it sometimes gave the final sentiments hidden within a sentence, for instance '... hence the sentence has Negative sentiment ...'. We used regex to find the sentimental labels. However, it sometimes failed for certain outputs. To fix this, we implemented a fallback mechanism, which search one of the 5 sentiment categories in the last two lines of output. With this mechanism, we were able to detect all the missed labels.

### 9.5 Takeaways

These results suggest that:

- **Implicit sentiment detection benefits substantially from structured reasoning scaffolds**.
- **Hybrid prompting compensates for the lack of fine-tuning on small datasets like SST**.
- **Multi-agent and contrastive setups improve pragmatic inference but increase generation variance**.

Our findings align with and extend prior work on pragmatic reasoning gaps in LLMs (Warstadt and Ryskina, 2023; Zhang et al., 2024).

## 10 Limitations

- **Inconsistent Output Formats:** Even with carefully structured prompts, LLMs sometimes deviated from the expected output, requiring robust post-processing and manual inspection.
- **Fragility of Regex-Based Extraction:** Extraction of sentiment labels relied on strict text patterns (e.g., "SENTIMENT: Positive"). Minor deviations (e.g., missing keywords, formatting errors) caused failures, highlighting the brittleness of regex parsing.
- **Domain Adaptation Needs:** Pre-trained LLMs, including Mistral, occasionally struggled with domain-specific sentiment expressions typical of movie reviews. Fine-tuning or few-shot adaptation would likely improve domain alignment.
- **Class Imbalance Impact:** The distribution of sentiment classes in SST is skewed, with far fewer examples for extreme sentiments ("Very Negative", "Very Positive"). This imbalance disproportionately affected F1 scores for minority classes.

- **Resource Constraints:** Due to limited GPU availability (e.g., T4x2 on Kaggle), full fine-tuning experiments were restricted to lightweight models. Larger-scale experiments (e.g., parameter-efficient tuning for Mistral or GPT-4o) were not feasible within the available compute budget.

- **Integration Mechanism Limitations** The debate-based integration approach using T5 may have introduced limitations:
  - The debate framework might overweight minority opinions
  - The T5 model may lack domain-specific knowledge to properly evaluate expert opinions
  - The integration may dilute strong signals from the Expert model

## 11 Future Work

While our experiments demonstrate the promise of hybrid prompting strategies for implicit sentiment analysis, several limitations remain:

- **Fine-Tuning:** Due to computational constraints, we were not able to perform full-scale fine-tuning of the Mistral-7B and GPT-4o model. Next step would be to fine-tune these models and evaluate again, hoping that our hybrid strategy will perform better. Future work could explore lightweight fine-tuning methods such as LoRA (Hu et al., 2021) or QLoRA (Dettmers et al., 2023) to improve domain adaptation without prohibitive compute costs.

- **Increasing the Dataset:** All experiments were conducted on the Stanford Sentiment Treebank (SST). Broader generalization remains untested. Extending evaluations to other datasets would validate the robustness of the prompting strategies across domains. In fact, creating a new comprehensive dataset which is never seen by LLMs would be a better way of evaluating their true performance.

- **Error Analysis Depth:** While we logged, categorized, and created fallbacks for parsing failures, deeper qualitative analysis (e.g., thematic clustering of misclassified examples, probing for biases in implicit reasoning) would provide richer insights into model behavior.

- **Multi-Agent Debate Scalability:** Although debate modeling improved performance, it also increased inference time and prompt length. Future research could explore more efficient aggregation mechanisms, such as weighted voting or reinforcement learning-based debate optimization.

- **Multi-LLM Debate Scalability:** Another exploration direction would be let different LLMs discuss a sentence and argue about the implied sentiment. This might possibly combine their intelligence and enhance the performance.

## 12 Contributions

- Maitri Patel: Led the design and implementation of multi-agent debate prompting and hybrid prompting strategies for the Mistral-7B model. Conducted all Mistral experiments, prompt engineering, model evaluation, and contributed substantially to report writing.

- Simran Khullar: Performed data preprocessing to classify the sentiments into categories and separated train, dev, and test set for training and testing purposes. Additionally, led the testing and evaluation of various prompting techniques on GPT-4o. Due to computational constraints, I mainly tested on GPT-4o mini.

- Zexuan Yang: Led fine-tuning three pretrained language models for sentiment classification using LoRA. Experiments demonstrated that fine-tuning would significantly boost model performance in our settings.

- Ao Xu: Implemented comprehensive test for fine-tuning classification LLMs. This included creating visualization tools that generated comparative loss curves, accuracy progression graphs, and precision-recall metrics across sentiment classes.

## 13 Conclusion

We investigated hybrid prompting strategies to improve implicit sentiment classification on several state-of-the-art models. Combining few-shot examples, role-play, contrastive reasoning, and debate modeling significantly improved over zero-shot baselines on the SST dataset. Our results show that

careful prompt engineering can enhance LLM reasoning in subtle sentiment tasks, even without full fine-tuning. Future work will explore lightweight fine-tuning methods and broader domain adaptation.

## References


Alex Warstadt and Maria Ryskina. 2023. The goldilocks of pragmatic understanding: Fine-tuning strategy matters for implicature resolution by LLMs. *In ACL Workshop on Figurative Language Processing.*

Xinyun Zhang, Jason Wei, Sebastian Gehrmann, and Dan Roth. 2024. Hybrid prompting: Towards reasoning-aware language models. *arXiv preprint arXiv:2501.08641*.

Veselin Kocijan, Vered Shwartz, and Ellie Pavlick. 2023. Large language models are not zero-shot communicators. *OpenReview*.

Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. Foundations and Trends® in information retrieval, 2(1–2), 1-135.

Zhang, X., & LeCun, Y. (2015). Text understanding from scratch. arXiv preprint arXiv:1502.01710.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers) (pp. 4171-4186).

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. OpenAI blog, 1(8), 9.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Sharan Narang, Adams Yu, Quoc V. Le, and Denny Zhou. 2022. Chain of Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, pages 24824–24837.

Jingye Liu, Swaroop Mishra, Yizhong Wang, Myle Ott, Sean Welleck, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. WANLI: Worker and AI collaboration for Natural Language Inference. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, pages 10267–10283.

Yilun Du, Yilun Ge, Shixiang Shane Song, Alex Tamkin, Vincent Y. F. Tan, Venkatakrishnan Gopalan, Rajarshi Das, Tatsunori Hashimoto, Percy Liang, and Yuhuai Wu. 2024. Improving Large Language Models with Multi-Agent Debate. In *International Conference on Learning Representations (ICLR 2024)*.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP 2013)*, pages 1631–1642.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, volume 1, issue 2, page 3.